

Practical Lab8: DNS Spoofing Forensics

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Course Code: SBT-DF203

Course Title : Basic Networking Skills for Digital Forensics

Lab Title: Lab 8 - DNS Spoofing Forensics

Instructor Name: Aminu Idris

Date: 24/9/2026

```
(kali㉿kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}

(kali㉿kali)-[~]
$ cd ~/SBT-DF203-Lab8

(kali㉿kali)-[~/SBT-DF203-Lab8]
$ pwd
/home/kali/SBT-DF203-Lab8

(kali㉿kali)-[~/SBT-DF203-Lab8]
$ find . -maxdepth 1 -type d -print
.
./exported
./evidence
./working
./screenshots
./scripts
./reports
```

```

(kali㉿kali)-[~/SBT-DF203-Lab8]
$ printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training
Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Afeez Bola
ji</p></body></html>\n' | sudo tee /var/www/html/index.html
[sudo] password for kali:
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an autho
rized simulation. Do not enter credentials.</p><p>Analyst: Afeez Bolaji</p></body></html>

(kali㉿kali)-[~/SBT-DF203-Lab8]
$ sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
994d555b6b808389b07c3c5715d90772a70bf07c539b0431a143ed0c818857da /var/www/html/index.html

```

```

(kali㉿kali)-[~/SBT-DF203-Lab8]
$ ip -br address | tee reports/interfaces.txt
lo                UNKNOWN          127.0.0.1/8 ::1/128
eth0              UP                192.168.159.128/24 fe80::bd40:b67:d601:4a1/64
docker0          DOWN            172.17.0.1/16

(kali㉿kali)-[~/SBT-DF203-Lab8]
$ ip route | tee reports/routes.txt
default via 192.168.159.2 dev eth0 proto dhcp src 192.168.159.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.159.0/24 dev eth0 proto kernel scope link src 192.168.159.128 metric 100

(kali㉿kali)-[~/SBT-DF203-Lab8]
$ ip neigh show | tee reports/arp_before.txt
192.168.159.254 dev eth0 lladdr 00:50:56:e1:5e:24 STALE
192.168.159.2 dev eth0 lladdr 00:50:56:f2:0e:73 REACHABLE

```

The chain-of-custody worksheet

Field	Entry
Case/lab identifier	SBT-DF203-Lab8-Afeez Bolaji Usman
Trainee name	Afeez Bolaji Usman
Date and time started	24/09/2026
Evidence file name(s)	dns_baseline.pcapng
Source or generation method	Self-generated on isolated Kali VM using network-namespaces victim simulation + dnsmasq legitimate resolver
Original SHA-256	e6990522e9623b9a7999efa7e1964ea7d95b4ab16d8c683a92a46b0514532c0a
Analysis workstation/VM	Kali Linux (kali㉿kali)
Notes on any changes	None — first capture attempt (dns_baseline2 timing race) discarded before evidence was finalized; final capture verified complete

Part A - Document the Baseline

```
(kali㉿kali)-[~/SBT-DF203-Lab8]
$ tshark -r evidence/dns_baseline.pcapng
  1 0.000000000 10.10.13.2 → 192.168.159.128 DNS 100 Standard query 0xa3bf A portal.icdfa.test
OPT
  2 0.000395575 192.168.159.128 → 10.10.13.2 DNS 104 Standard query response 0xa3bf A portal.i
cdfa.test A 192.168.159.128 OPT
  3 0.041223899 10.10.13.2 → 192.168.159.128 DNS 77 Standard query 0x864f A portal.icdfa.test
  4 0.041312908 10.10.13.2 → 192.168.159.128 DNS 77 Standard query 0x44ce AAAA portal.icdfa.te
st
  5 0.041349236 192.168.159.128 → 10.10.13.2 DNS 93 Standard query response 0x864f A portal.ic
dfa.test A 192.168.159.128
  6 0.041453712 192.168.159.128 → 10.10.13.2 DNS 77 Standard query response 0x44ce Refused AAA
A portal.icdfa.test
  7 0.042100982 10.10.13.2 → 192.168.159.128 DNS 77 Standard query 0x44ce AAAA portal.icdfa.te
st
  8 0.042257564 192.168.159.128 → 10.10.13.2 DNS 77 Standard query response 0x44ce Refused AAA
A portal.icdfa.test
  9 0.045177807 10.10.13.2 → 192.168.159.128 TCP 74 44748 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS
=1460 SACK_PERM TSval=3226257159 TSecr=0 WS=512
 10 0.045207663 192.168.159.128 → 10.10.13.2 TCP 74 80 → 44748 [SYN, ACK] Seq=0 Ack=1 Win=6516
0 Len=0 MSS=1460 SACK_PERM TSval=303889879 TSecr=3226257159 WS=512
 11 0.045221025 10.10.13.2 → 192.168.159.128 TCP 66 44748 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len
=0 TSval=3226257159 TSecr=303889879
 12 0.045287670 10.10.13.2 → 192.168.159.128 HTTP 147 GET / HTTP/1.1
 13 0.045301527 192.168.159.128 → 10.10.13.2 TCP 66 80 → 44748 [ACK] Seq=1 Ack=82 Win=65536 Le
n=0 TSval=303889879 TSecr=3226257159
 14 0.047916882 192.168.159.128 → 10.10.13.2 HTTP 521 HTTP/1.1 200 OK (text/html)
 15 0.047950047 10.10.13.2 → 192.168.159.128 TCP 66 44748 → 80 [ACK] Seq=82 Ack=456 Win=64512
Len=0 TSval=3226257162 TSecr=303889882
 16 0.048499928 10.10.13.2 → 192.168.159.128 TCP 66 44748 → 80 [FIN, ACK] Seq=82 Ack=456 Win=6
4512 Len=0 TSval=3226257162 TSecr=303889882
 17 0.048854682 192.168.159.128 → 10.10.13.2 TCP 66 80 → 44748 [FIN, ACK] Seq=456 Ack=83 Win=6
5536 Len=0 TSval=303889883 TSecr=3226257162
 18 0.048871020 10.10.13.2 → 192.168.159.128 TCP 66 44748 → 80 [ACK] Seq=83 Ack=457 Win=64512
Len=0 TSval=3226257163 TSecr=303889883
 19 5.051781362 76:b0:2f:fc:15:6d → c6:8b:cf:9a:fe:b4 ARP 42 Who has 10.10.13.1? Tell 10.10.13.2
 20 5.052080728 c6:8b:cf:9a:fe:b4 → 76:b0:2f:fc:15:6d ARP 42 10.10.13.1 is at c6:8b:cf:9a:fe:b4
```

```
(kali㉿kali)-[~/SBT-DF203-Lab8]
$ cat ~/SBT-DF203-Lab8/reports/training_page_sha256.txt
994d555b6b808389b07c3c5715d90772a70bf07c539b0431a143ed0c818857da /var/www/html/index.html

(kali㉿kali)-[~/SBT-DF203-Lab8]
$
```

The training page hash, already captured:

994d555b6b808389b07c3c5715d90772a70bf07c539b0431a143ed0c818857da

Field	Value
Victim IP/MAC	10.10.13.2 / 76:b0:2f:fc:15:6d
Analyst IP/MAC	192.168.159.128 / c6:8b:cf:9a:fe:b4

Gateway IP/MAC	192.168.159.128 (your dnsmasq instance, acting as legitimate resolver for this training domain)
DNS resolver IP	192.168.159.128 (your dnsmasq instance, acting as legitimate resolver for this training domain)
Legitimate answer	portal.icdfa.test → 192.168.159.128
Responder IP/MAC	192.168.159.128 / c6:8b:cf:9a:fe:b4 (same as analyst — correct, dnsmasq runs there)
TTL	0
Query-response time	0.4ms